

Synaptic Plasticity as Short-Term Memory

Companion to the interactive explainer “The synapse that never fills up” · audience: a data scientist meeting this concept for the first time

The concept, and why it is timely

Softmax attention answers a query by comparing it against a stored key–value pair for every prior token. That store — the KV cache — grows linearly with sequence length and is the dominant memory and latency cost of long-context Transformers. **Synaptic plasticity as short-term memory** is the alternative recurring across recent architectures: instead of keeping every token, write each association into a *single fixed-size weight matrix* using a Hebbian rule — units that co-activate strengthen their connection — and read it back with a cue. Memory becomes a property of the wiring, updated as the model reads, rather than a growing list of past states.

The mechanism, precisely

Remove the softmax and the attention sum reassociates: $\text{out}_i = \sum (q_i \cdot k_i) v_i = q_i^T (\sum k_i v_i^T) = q_i^T S$. The state S is a fixed $d_k \times d_v$ matrix with a recurrent update — **write** $S \leftarrow \lambda S + k v^T$ (a Hebbian outer product), **read** $r = S^T q$. This is the *fast-weight / linear-attention* view of associative memory.

The technical change from a Transformer: state size is decoupled from sequence length. The trade-off: reading a cue k_j yields $v_j (k_j \cdot k_j) + \sum_{i \neq j} v_i (k_i \cdot k_j)$ — the intended value plus **crosstalk** from every other memory overlapping the cue. Nothing is evicted; recall fails when accumulated crosstalk outgrows the signal. Capacity is set by d_k and by how near-orthogonal the keys are, **not** by a slot count.

Where BDH and BDH-CQ fit

Dragon Hatchling (BDH) [1] makes this the architecture rather than an add-on: attention is reformulated as synaptic memory over a scale-free network of locally interacting neurons, so reasoning and memory share one computational fabric. Its GPU form, BDH-GPU, is built from ReLU-low-rank transformations with linear attention — importantly *not* a Mamba-style SSM. Two reported properties bear directly on the mechanism above: roughly **5% of neurons are active**, and activations are **non-negative**. Non-negativity removes sign cancellation, so crosstalk that averaged out for signed keys now always adds; sparsity is what keeps that linear growth survivable.

BDH-CQ [2] is a later family member for in-context reasoning; it relates its contextual memory to the same fast-weight / linear-attention view of association, with a special case where state accumulates additively per demonstration. BDH's role here is central; BDH-CQ's is a supporting reference, and we do not inflate it.

Positioning: how the fixed-state view compares on what actually matters

Approach	State vs. length	Interference	Interpretability	Note
Softmax Transformer (KV cache)	grows linearly	none (exact recall)	low	strong recall, costly at length
Linear attention / fast weights [3]	fixed	crosstalk-limited	medium	the substrate our artifact builds
State-space models (Mamba)	fixed	selective decay	medium	gated, not Hebbian; BDH is not this
BDH / BDH-CQ [1,2]	fixed, synaptic	sparsity-mitigated	high (monosemantic synapses)	reasoning + memory unified

Primary sources. [1] Kosowski et al., *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*, arXiv:2509.26507 (2025). [2] Pathway, *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888 (2026). [3] Katharopoulos et al., *Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention*, arXiv:2006.16236 (2020).

AI assistance was used for drafting; every sentence, number, and citation is defensible and reproducible via `scripts/verify.mjs`. Artifact and full disclosure in the repository README.

Evidence, correctly labelled

Our interactive artifact *demonstrates the mechanism* on a toy associative memory with random keys, and every number is reproducible (node `scripts/verify.mjs`, seed 42): state stays constant while the KV cache grows; recall decays smoothly with load (100%→77% from 2→48 pairs at $d_k=32$); measured key-overlap matches theory (dense $\approx \sqrt{(2/\pi)}/\sqrt{d_k}$; sparse $\approx a/d_k$).

We did **not** run any BDH checkpoint. Pathway's headline results — BDH's ~97.4% solve rate on Sudoku Extreme without chain-of-thought, and BDH-CQ's 29.5% pass@2 on public ARC-AGI-1 at ~\$0.0007/task — are **developer-reported benchmarks** cited from primary sources, not independent reproductions or deployments. BDH's pretraining-scaling from 1B toward 600B parameters is an early reported experiment.

Strengths, and the most important limitation

The fixed-state view buys constant memory and cheap updates, and — in BDH's case — an interpretable substrate where synapses can encode identifiable concepts. The honest limitation: **within-session synaptic memory is not durable cross-session learning**. The matrix resets when context ends; consolidating useful fast state into slow trained weights is an open problem BDH frames rather than solves. The open question worth watching is whether learned sparse synaptic codes keep crosstalk manageable at production scale, where our random-key substrate is deliberately pessimistic.